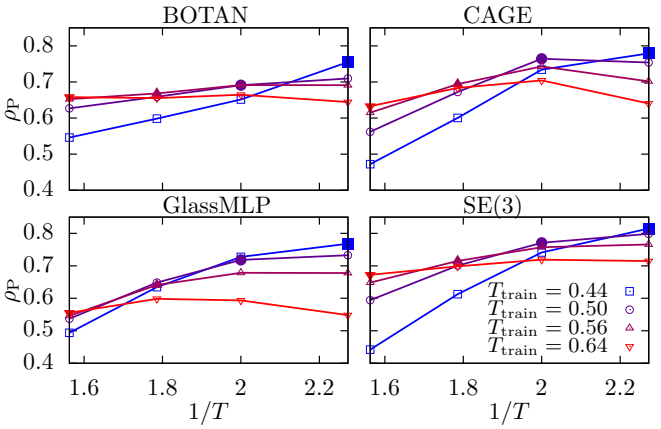

FIG. 5. **Benchmark task (B)** Transferability in temperature T of trained networks in the KA system (3D) at the structural relaxation time τ_α. Each network is trained at $T_{\rm train}$ (marked by the full symbol) and applied to all four different temperatures (marked by open symbols). The color codes for $T_{\rm train}$ which changes smoothly from red ($T = 0.64$, high temperature) to blue ($T = 0.44$, low temperature).

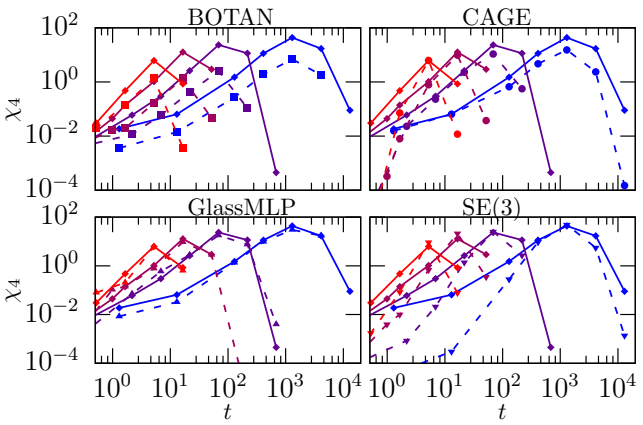

FIG. 6. **Benchmark task (C)** Dynamic susceptibility $\chi_4(t)$ for different temperatures ($T = 0.44$, 0.5, 0.56, 0.64), as predicted from ML techniques (dashed lines), compared to the ground truth (full lines) in the KA system (3D). The color code is as in Fig. 5.

dynamics relationships learnt at a given temperature to make predictions at a different temperature. The results are actually quite remarkable as transferability is generically quite good for all models. This shows that these relationships only evolve quite smoothly across the range of temperatures investigated here. In particular, the models trained at $1/T = 2.0$ ($\tau_\alpha = 210$) perform nearly as good in predicting propensity at $1/T = 2.25$ ($\tau_\alpha = 4100$) as the models trained directly on $1/T = 2.25$. The SE(3) method seems to be particularly suited to transfer to lower temperatures, opening the possibility to study structural relaxation at much lower temperatures.

Task C. Finally, we investigate the performance of the models to predict the correct extent of dynamic heterogeneity. A time-dependent scalar that quantifies heterogeneities is the dynamic susceptibility

$$\chi_4(t) = N\left(\langle \bar{C}^2_{\mathcal{R}}(t)\rangle - \langle \bar{C}_{\mathcal{R}}(t)\rangle^2\right) \tag{2}$$

calculated from the system-averaged overlap function $\bar{C}_{\mathcal{R}}(t) = (1/N)\sum_{i\in N}\Theta\left(0.3 - \mathcal{R}_i(t)\right)$. Here, N is the number of particles in the system, and $\Theta(x)$ the Heaviside function. This definition separates particles into active ($\mathcal{R}_i > 0.3$) and frozen ($\mathcal{R}_i \leq 0.3$). The threshold value of 0.3 is a common choice [121, 122] and corresponds to values slightly larger than the plateau in the mean-squared displacement [121], implying that particles identified as active have typically left their initial cages. In Fig. 6, we compare the results of the predictions to the ground truth MD simulations. Despite the overall good performance in the Pearson correlation, we now observe strong differences between the various techniques. For example, the improvement in performance of SE(3) compared to BOTAN shown in Fig. 4 can be connected to their different learning of the correct dynamic heterogeneity. In a recent attention-based GNN extension, this heterogeneity has been explicitly targeted during the training procedure to improve the performance of the deep network [123]. The best overall performance in predicting $\chi_4(t)$ is achieved by GlassMLP, which was specifically constructed to learn and predict dynamic heterogeneity [30]. This analysis shows that the Pearson correlation is not entirely sufficient to quantify the performance of a model. Additional dynamical observables, such as the dynamic susceptibility $\chi_4(t)$, should be investigated to better characterize the ability of models to realistically describe structural relaxation in glass-forming liquids.

In the SM [54] we provide further benchmarking, by investigating: (a) the coefficient of determination R^2, another popular measure to study the performance of ML models; (b) the probability distributions of the predicted propensities, which correspond to a second contribution to dynamic heterogeneity; (c,d) the scatter plots and the snapshots directly comparing the true and predicted propensities; (e) the cross-correlations between the different structural descriptors and the models investigated in this section. We also provide additional information on (f) learning curves, (g) transferability to lower temperatures $T = 0.4$, and (h) bond-breaking propensity.

Another conclusion of additional benchmarks, which goes beyond the scope of this roadmap, is that correlation coefficients display some system dependence [73]. Hard sphere glasses and systems with strong icosahedral order, for example, show systematically higher correlation coefficients between structural and dynamical descriptors than Kob-Andersen mixtures [24, 51]. The same statement holds true for different dynamical descriptors and coarse-grained quantities. We therefore strongly encourage the use of identical datasets and labels (i.e., dynamical descriptors) to enable comparability.

Among possible extensions of GlassBench, it would

be particularly worth including a diverse set of models of glass-formers: fragile molecular glass-formers, strong network-forming glasses and potentially metallic glasses. The first step in creating a new dataset is the sampling of independent configurations at a given temperature or density. This step could leverage enhanced sampling techniques such as parallel tempering [124] or swap Monte Carlo [125]. Subsequently, for each structure, molecular dynamics or ab-initio dynamics simulations will have to be performed to study the structural relaxation of the systems and calculate the isoconfigurational average [87].

It would also be interesting to broaden the tasks. For example, other ML models, such as the softness derived from SVC, target predicting local energy barriers for short-time rearrangements instead of long-time structural relaxation [19, 85] (see Sec. IV). For this class of ML techniques, it would be preferable that the descriptor can effectively separate the particles by their probability of rearrangement, $P_R(S)$. To encourage further development in this area, it would be desirable to perform a similar benchmarking and investigate whether modern ML models can outperform the state of the art.

Similarly, there is much interest in understanding plastic events and failure of glassy materials under external load. Contrarily to equilibrium structural relaxation, this protocol takes place far from equilibrium, but displays similar characteristics. Recently, this field has been reviewed and benchmarked in a collaborative publication [11]. The focus, however, was not yet on advanced ML techniques, which further demonstrates the very quick development of this field. It would be interesting to analyse whether the techniques presented in this roadmap can help investigating glass deformation under shear [21, 126–128].

VI. SUMMARY AND FURTHER DIRECTIONS

One of the main challenges in understanding glass-forming liquids is connecting their structural properties to emergent relaxation dynamics and plastic deformation. Since the amorphous structure lacks apparent long-range order, machine learning techniques are potentially useful to extract important information from simulations or experimental measurements and eventually generate new data. We therefore anticipate that machine learning will have a decisive influence on the field of glass physics in the upcoming years.

We have given both an overview of recent achievements in using machine learning to advance our fundamental understanding of glass physics, and discussed important research questions for the future. We identified three main fields in which machine learning could impact research on glass physics: (i) automatic feature detection to better understand locally favored amorphous structures, (ii) forecasting of future microscopic dynamics and investigation of dynamical heterogeneities, and (iii) the construction of data-driven effective models of glassy dynamics. For each of these fields we have proposed concrete methodology that could be used to answer some of the identified research questions. Finally, we have provided a dataset and benchmarking to provide a common ground to compare ML methods for glassy dynamics and enable reproducibility. This work will hopefully provide inspiration and guidelines to researchers on how to further develop the field with the ultimate goal to better understand universal properties of glass-forming liquids and, eventually, the glass transition itself.

The number of potential applications of ML techniques for studies of glass-forming liquids and glasses goes of course much further than what we covered in this article, focusing on fundamental aspects of glassy dynamics. This includes investigation of specific material properties [69, 70, 129, 130] or material discovery [67, 131, 132], machine learning force fields [133], and particle identification in experimental data [134, 135]. Furthermore, the connection between ML and glassy physics has also gone in the reverse direction, by borrowing methods developed for the study of disordered systems to analyze central theoretical ML problems. In fact, the connection between the rough energy landscape of amorphous materials and optimization defined by loss functions with many local minima has been used to better understand and optimize learning of neural networks [136–141].

We anticipate that modern ML frameworks will continue to impact glass research and lead to the development of new major directions in the field. We close this roadmap by discussing exciting new concepts that have the potential to play an important role in future research.

A. Attention and transformers

An important advance in recent ML architectures is the "attention mechanism" [142]. Using it in ML methods for glassy dynamics has a lot of potential. The fundamental concept behind attention is to assign a learnable level of importance to specific parts of the input or intermediate representation. This could be distinct words in sentences (see, e.g., ChatGPT), amorphous structures' specific features, channels in a deep representation or neighboring atoms in a graph representation. Broadly speaking, this can be achieved by making learned weights themselves dependent on the input.

Salient examples of successful applications of attention are AlphaFold v2 and RoseTTAFold [143, 144], which both use a rotation-equivariant attention-based transformer to predict the three-dimensional structure of proteins. This architecture differs from the graph network architectures discussed earlier by the dense character of its computation mechanism: all atoms in the input can exchange information with all other atoms (with a learned modulation as a function of the distance), allowing for a more flexible computation. Variants of these architectures tend to obtain results which are competitive

with sparse graph networks on benchmark tasks [145]. Recently, a network with a self-attention mechanism was developed for predicting glassy dynamics [123], showing that the curse of overfitting can be avoided. The broader concept of input-dependent weights was also used in a similar context to learn dynamic heterogeneities over a wide range of temperatures at which attention must be paid to temperature-dependent length scales [89]. Both of the approaches are, however, rather direct applications of the idea of positional encoding for learning attention weights, and more complex networks, such as full-fledged transformers [142] are expected to be used in future research.

Going in the opposite direction from transformers, attention mechanisms can also be used to make models more interpretable. For example, they can help identify the task-relevant sectors in the input data [146]. Additionally, attention can also be incorporated into other architectures, for example in combination with a temporal encoding in time-series forecasting [147], to identify relevant parts of past trajectories.

B. Self-supervised, semi-supervised and reinforcement learning

So far, we have mainly focused on "traditional" unsupervised and supervised learning techniques since they are better established in the field. However, there are several other learning paradigms which are potentially useful for future projects, some of which have started to be used in glass research.

To learn the connection between structural order and structural relaxation in a way that reconciles the unsupervised and supervised approaches, a possibility is to use semi-supervised learning [148]. Concretely, the idea is to perform self-supervised learning using only unlabeled input configurations by designing a pretext task, such as reinserting a particle which has been artificially removed from a configuration, denoising particles' positions, or predicting local quantities (such as the potential energy of each particle or its distance to its quenched position) [91]. Once a representation has been learned to perform this mock task, one can fine-tune only a handful of parameters to correlate the learned representation to the relevant dynamical variable. Such self-supervised pretraining has proven effective in increasing performance for various downstream tasks that are similar in spirit to glassy dynamics prediction, such as molecular properties [149], crystalline material properties [150] or organic semiconductors optoelectronic properties prediction [151]. In the application to glasses, the key element of this approach is to build most of the network without looking at labels, so the output of such a method may be more acceptable as a bona fide structural descriptor, as opposed to heavy networks relying purely on supervised learning. Additionally, it requires fewer labeled data which becomes very important at low temperatures at which sampling becomes difficult. The described methodology could therefore also be used to improve transferability of pre-trained models.

Although reinforcement learning is a well established tool in the field of machine learning [152] only very recently it has found applications in physics for improved sampling [153, 154] or structure optimization [155, 156]. The general idea behind reinforcement learning is to learn taking specific actions when reaching certain states. The goal is to find the policy of actions leading to optimal results, as quantified by a reward function. Applying this approach to the example of searching ground states in spin glasses, the state would be the observed structure, action would be a spin-flip and reward the energy change after several spin flips [154]. Along these lines, Bojesen [157] has formulated the Metropolis-Hastings algorithm in a reinforcement learning setting suitable for simulations of spin systems. Recent work by Galliano *et al.* has extended this approach to learn novel Monte Carlo moves that accelerate sampling of supercooled liquids [158], establishing connections with related adaptive Monte Carlo methods [159, 160]. One of the goals of this line of research is to improve and generalize the swap Monte Carlo algorithm, which demonstrated an impressive performance for specifically adapted glass models [125, 161] and has led to a series of exciting new insights into supercooled liquids [106, 117]. Devising general-purpose enhanced sampling algorithms to simulate glassy systems represents an exciting challenge for future research [162].

C. Generative models

Another promising emerging direction for applying ML techniques to problems relevant to fundamental aspects of glass physics is the use of generative models (GM). One of the key problems in theoretical studies of glassy dynamics is that of sampling. In fact, sampling efficiently configurations x_i from a Gibbs distribution of the form $P(x_i) = \exp[-\beta U(x_i)]/Z$, where $\beta^{-1} = k_B T$ is the target inverse temperature, and $U(x_i)$ is the known potential energy is very challenging for systems exhibiting glassy dynamics. Different from generative modelling of images, in which one estimates an unknown probability distribution from data, here the target distribution is therefore known from the start and sampling from it is the challenge.

A first line of research was proposed independently by Noé *et al.* under the name of Boltzmann generators [163] and by Wu *et al.* [164] using variational autoregressive networks. The idea is to consider a much simpler distribution $\hat{P}(z)$, such that one can easily sample z_i in a single shot. This can, for example, be a Gaussian model, an autoregressive model, or a Gibbs distribution at very high temperatures at which we can sample efficiently. After learning, independent samples x can then be generated using the invertible map $x_i = f(z_i)$ of the model. By